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Research Interests

Program analysis and software testing; AI for software engineering, particularly evaluating and validating LLM-generated code changes; security tooling for production systems (fuzzing, static analysis); data-intensive and high-performance computing infrastructure.

Education

Arizona State University

Aug 2022 – May 2024

M.S., Computer Science GPA 4.00/4.00

Tempe, AZ

- Coursework: Foundations of Machine Learning, Artificial Intelligence, Knowledge Representation, Cloud Computing, Data Visualization, Software Project Management.

Motilal Nehru National Institute of Technology (MNNIT)

Jun 2014 – May 2018

B.Tech., Computer Science & Engineering

Prayagraj, India

Publications

1. Powell, B., Young, S.E., Sperling, J., Duvall, A.E., Shrestha, A. “Where are all the Electric Vans? Electrification Considerations for On-Demand Transit.” ASCE International Conference on Transportation and Development (ICTD), 2025.
2. Powell, B., Henao, A., Young, S., Sperling, J., Duvall, A., Shrestha, A. “The ABCs of On-Demand Transit (ODT).” Conference presentation, National Laboratory of the Rockies, 2026. Report NLR/PR-5400-94851. doi:10.2172/3030031
3. Nikhil, D.V., Dewang, R.K., Sundar, B., Agrawal, A., Shrestha, A.N., Tiwari, A., Mewada, A. “Predicting Native Language with Machine Learning: An Automated Approach.” In Artificial Intelligence Techniques in Power Systems Operations and Analysis, pp. 195–206, CRC Press, 2023. doi:10.1201/9781003301820-11
4. Nikhil, D.V., Dewang, R.K., Sundar, B., Agrawal, A., Shrestha, A.N., Tiwari, A. “An Android Application for Automatic Content Summarization of News Articles Using Multilayer Perceptron.” Proceedings of ICBMA 2019, Lecture Notes in Networks and Systems, vol. 150, pp. 379–394, Springer, 2021. doi:10.1007/978-981-15-8377-3_32

Research Experience

Research Software Engineer Intern

Aug 2023 – Sep 2025

National Renewable Energy Laboratory (NREL)

Golden, CO

- Built and validated a 1 TB data-processing pipeline for the national EV infrastructure grid simulation on HPC (Python, Polars, Pydantic, Parquet), reducing execution time from 10 hours to 30 minutes. Architected a Pydantic validation layer enforcing strict configuration schemas.
- Made the OpenPATH public dashboard testable. Its chart logic called `get_ipython().run_line_magic()` at import time, so no test could import the module. Decoupled the IPython dependency to be runtime-detected and optional, then built the CI harness that did not previously exist (GitHub Actions → docker-compose → pytest with coverage), replacing a five-step manual verification ritual. Design tradeoffs documented publicly in em-public-dashboard#171, including why notebook-level testing was rejected and why automated chart-image comparison was scoped out as unsolved.
- Redesigned the OpenPATH public dashboard (Python, JavaScript, MongoDB) serving 5,000+ users across 34 deployments; replaced pie charts with 100% stacked bar charts for cross-program comparability.
- Contributed to RouteE Compass, an open-source energy-aware routing engine in Rust enabling Google Maps’ eco-routing. Recognized with the U.S. DOE Vehicle Technologies Office AMR Team Award, 2025.
- Built a machine-learning pipeline (random forest with feature engineering and cross-validation) for deployment-stage commute-mode inference over large-scale mobility datasets.
- Teams: e-mission/OpenPATH, RouteE Compass, EVI-Pro, EVI-Grid National Framework, On-Demand Transit Research.

Project Team Leader (graduate research project)

Jan 2023 – May 2023

Arizona State University

Tempe, AZ

- Led a team of seven graduate students on information assurance and security in machine learning, targeting malicious URL detection in social networks; investigated DCNN and CNN approaches.

Open-Source Contributions

- e-mission/em-public-dashboard: 33 pull requests authored, 22 merged, covering chart, metrics and colour work plus the test-automation effort described above.

- NatLabRockies/routee-compass (Rust): four merged pull requests, the largest (#286) moving JSON getter extensions into the core crate across 40 files; also SI energy-rate units (#280), a TOML inject format (#285), and gzip-compressed query files (#483). Plus conda-forge feedstock #14 for osx-arm64.
- nanvix/nanvix (Rust): six commits merged to the default branch of Microsoft Research's multikernel OS for serverless sandboxing: error-type definitions in the kernel `sys` layer and build-script hardening.

Industry Experience

Software Engineer Dec 2025 – Present
 Exito Tech Inc. New York, NY

- Building an AI chatbot on AWS using Anthropic Claude and retrieval-augmented generation to improve response grounding and quality.

Software Engineer Intern II May 2023 – Aug 2023
 Yahoo Inc., Edge Infrastructure Champaign, IL

- Integrated four fuzzing engines (Google FuzzTest, libFuzzer, AFL++, Honggfuzz) into Yahoo's large-scale production DNS infrastructure in C++17, identifying and reporting two critical memory-management vulnerabilities.
- Automated the fuzzing workflow with CI/CD pipelines (ScrewDriver, Makefile, CMake), evaluating enterprise Linux production images before deploying on an Ubuntu base image.
- Authored a ~50-page report distilling research on coverage-guided fuzzing engines: the available integration methods, the tradeoffs of each, and how to reproduce the setup unaided.

Software Engineer I & II Jun 2018 – Jun 2022
 Philips India Ltd., Ultrasound R&D Bangalore, India

- Architected a multi-threaded, low-latency (5 ms) real-time audio processing module (Swift, C++, AVFoundation/AVAudioEngine) for Pulse Wave Doppler playback on the Lumify iOS app, under an FDA-regulated release process.
- Rhythm Service Platform (C++, MVVM): designed multi-level user login using a publish-subscribe pattern and layered architecture, maintaining >95% platform-layer and >85% application-layer unit test coverage.
- Triaged and resolved critical multi-threading and connectivity defects in a 20 GB legacy C++ codebase; built a C++/C# static-analysis CLI tool that identified stale parameters, resolving ten high-priority production incidents.
- Designed a secure multi-level authentication portal (C++, C#, WPF/MVVM) incorporating AES and SHA-256. Shipped ultrasound features (Quad AFI, High Q Doppler, 3D Preset within Preset) reducing exam execution time by up to 40%.

Selected Projects

- Energy Policy RAG (github.com/iantei/energy-rags) : retrieval-augmented generation system with a Medallion (bronze/silver/gold) ETL on Polars and Parquet, Docling PDF parsing, ChromaDB, LangChain, and Dockerized GPU inference via Ollama; 18-test integration suite.
- Realtime Audio Lab (github.com/iantei/swiftui-realtime-audio) : lock-free single-producer/single-consumer ring buffer with acquire/release atomics and wait-free parameter passing, so the CoreAudio render thread never allocates or locks under a ~5.3 ms deadline (48 kHz / 256-frame buffer). Windowed real FFT via Accelerate/vDSP; Swift 6 strict concurrency. Tests cover ring-buffer wrap-around, torn reads, and oscillator phase continuity. Runs as a macOS host application.
- Distributed Banking System: client-centric consistency models (Read-Your-Writes, Monotonic Writes) and Lamport logical clocks for causal event ordering across replicated servers (Python, gRPC).

Technical Skills

Languages: Python, C++, Rust, Swift, Java, C#, JavaScript, SQL, C

Analysis & Security: fuzzing (Google FuzzTest, libFuzzer, AFL++, Honggfuzz), static analysis, vulnerability identification, AES/SHA-256

Data & HPC: Polars, Parquet, PostgreSQL, MongoDB, ChromaDB, Slurm, distributed transactions and consistency

ML & GenAI: scikit-learn, PyTorch, Anthropic API, LangChain, RAG architecture, Ollama

Tooling: Docker, Docker Compose, GitHub Actions (CI/CD), pytest, CMake, AWS (EC2, S3, SQS, Lambda, IAM, ECR)

Awards and Honors

- U.S. Department of Energy, Vehicle Technologies Office: AMR Team Award, 2025 (RouteE & FASTSim, enabling Google Maps eco-routing).
- Philips Ultrasound Hackathon: Winner 2019–20 (High Q Doppler Trace Enabler, 200 participants); Runner-up 2018–19 (Unified Defect Writing Tool, 150 participants).
- Philips Planning Excellence Award: Winner, Q4 2021, cited for quality, predictability and productivity; Team Winner, Q1 2021; Q3 2022.

References

Available on request.